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Sensor Fusion for Methane Emission Quantification

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Abstract

Methane emissions pose significant environmental and safety risks, necessitating accurate and reliable detection methods. This paper presents a novel approach to methane emission quantification through sensor fusion, integrating data from multiple gas leakage sensors and robotic platforms. By combining diverse sensor modalities, our method overcomes the limitations of traditional single-sensor systems, enhancing detection precision and robustness.

We employ autonomous data collection using robotics, followed by advanced data processing through machine learning and computer vision techniques. Experimental validation involved testing various sensor configurations under controlled leak scenarios to identify the optimal fusion strategy. Comparative analysis against conventional methods demonstrated superior performance in both detection sensitivity and quantification accuracy, particularly at low emission rates.

Our findings highlight the potential of sensor fusion to revolutionize methane monitoring by enabling earlier and more reliable leak detection. This integrated approach offers a scalable, field-deployable solution for emission management, contributing to safer operations and reduced environmental impact in the energy sector.

Introduction

Methane is a highly potent greenhouse gas, with a global warming potential significantly greater than that of carbon dioxide over a 20-year period. Its unintentional release into the atmosphere, particularly from industrial sources such as oil and gas infrastructure, poses serious environmental and safety risks. Accurate detection and quantification of methane emissions are therefore critical for both regulatory compliance and environmental stewardship. However, traditional methane monitoring systems often rely on single-sensor technologies, which can suffer from limitations in sensitivity, spatial resolution, and robustness under varying environmental conditions. These constraints can lead to delayed detection, underestimation of leak severity, and increased operational risks.

In response to these challenges, this paper introduces a novel approach to methane emission quantification based on sensor fusion. By integrating data from multiple types of gas leakage sensors—each with distinct detection principles and sensitivities—our method aims to overcome the shortcomings of single-sensor

systems. The fusion of heterogeneous sensor data enables a more comprehensive and accurate representation of methane concentration and dispersion patterns. Furthermore, the use of autonomous robotic platforms for data collection enhances spatial coverage and repeatability, while reducing human exposure to potentially hazardous environments.

This research explores the design and implementation of a sensor fusion framework that combines real-time data from various sensors, then theorize an algorithm for quantification that uses advanced machine learning and computer vision techniques. We then proceed to test the hardware infrastructure with two real Optical gas imaging (OGI) cameras.

Ultimately, this work contributes to the development of more effective methane monitoring solutions, supporting the energy sector's transition toward safer and more sustainable practices. As a next step, the integration of sensor fusion and robotics may represent a significant step forward in emission management, offering a scalable and adaptable tool for real-world deployment.

Literature Review

Traditional methods for detecting methane leaks, such as handheld detectors and onsite inspections, are labor-intensive and provide limited spatial coverage. Over the past decade, advanced sensing techniques have emerged to enhance detection frequency and accuracy. Innovations like remote sensing methods (e.g., infrared cameras, laser-based sensors on vehicles or aircraft) and autonomous platforms are revolutionizing how methane leaks are detected and quantified. These technologies aim to increase precision and provide quantitative emission estimates in real-world energy applications.

However, no single sensor type can robustly detect and quantify methane leaks in all scenarios. Sensor fusion, which combines complementary sensing modalities, leverages their strengths and mitigates individual weaknesses. Key sensor types include electrochemical/catalytic sensors, which are compact and sensitive but have limited range, and optical gas sensors like tunable diode laser absorption spectroscopy (TDLAS) and cavity-enhanced spectrometers, which offer high sensitivity and fast response over longer distances. OGI using infrared cameras visualizes methane plumes but is qualitative and requires human interpretation.

By integrating multiple sensors, a system can compensate for individual blind spots. For example, a fixed sensor node might combine a fast optical sensor with a local catalytic sensor, or a robotic platform might carry both a gas detector and a thermal camera. Advances in sensor miniaturization and embedded electronics enable drones and small robots to carry diverse sensor payloads, gathering co-registered data that provide a more reliable picture of methane leaks. This fused data can be processed in real-time to improve detection confidence and refine emission quantification, making multi-sensor fusion a powerful tool for accurate methane leak detection.

Multi-camera optical gas imaging (OGI) fusion techniques have rapidly evolved to enable quantitative methane-emission assessments in oil & gas settings. [Rangel et al. \(2020\)](#) first demonstrated two synchronized IR OGI cameras, applying variational optical-flow and stereo-disparity algorithms to recover dense 3D "scene-flow" velocity fields of methane plumes; field validation at a biogas plant showed that the computed 3D velocities closely matched actual plume motions, establishing the feasibility of dual-camera fusion for leak-rate estimation. Building on this, [Rangel et al. \(2021\)](#) introduced a catadioptric-stereo configuration in which a single OGI camera views the plume via planar mirrors to synthesize two stereo perspectives; this approach minimizes temporal and radiometric misalignments inherent in dual-camera setups and delivers more consistent 3D velocity estimates than conventional rigs.

Subsequent work has both refined single-camera quantitative OGI (QOGI) and pushed the boundaries of multi-view fusion. [Ilonze et al. \(2024\)](#) evaluated the FLIR QL320 QOGI system across controlled releases in near-field conditions, revealing that rigorous calibration yields rapid leak-rate measurements but that accuracy degrades for low-flow leaks (<1 g/s) absent multi-angle corrections. [Bergau et al. \(2024\)](#)

advanced active OGI mass-flow quantification (qOGI) by integrating controlled IR illumination, sensitivity adaptation, and camera-to-gas-distance sensing; in biogas-plant trials, their system achieved mass-flow estimates within 10 % of reference values for leaks above 0.1 kg/h, illustrating how fusion of active illumination and multi-view data improves quantification.

Spectral-fusion concepts have also emerged: Wang et al. (2025) presented a multi-channel OGI (MOGI) sensor combining two absorption-band filters and one reference channel on a single camera to extract concentration-path-length information without stereo geometry, achieving <15 % error in controlled methane releases. On the theoretical side, Kang & Zimmerle (2022) surveyed radiative-transfer modeling and inversion methods for passive IR gas imaging, arguing that fusion across temporal, angular, or spectral OGI frames is essential to overcome the insufficiency of single-pixel radiance for reliable quantification. Early foundational work by Watremez et al. (2016) applied 3D reconstruction algorithms to infrared gas-imaging data in SPE field tests, validating the core concept of volumetric gas-flow estimation from multi-view IR imagery. More recently, Zhang et al. (2025) demonstrated rapid 3D reconstruction of leaking gas clouds using two multi-spectral imaging systems and inverse-Abel transforms, underscoring the versatility of multi-sensor fusion strategies for real-time plume mapping.

Together, these studies chart a clear trajectory: from dual-camera stereo OGI to mirror-based single-camera fusion, active-illumination enhancements, and spectral-channel integration, all pointing toward scalable, high-accuracy methane quantification systems for oil & gas infrastructure.

On the other hand, our study focuses more on the hardware integration of multiple OGI cameras to deploy an infrastructure for concurrent data acquisition. Moreover, we propose also a novel algorithm for sensor fusion and 3D flow quantification. For this research we focus just on multiple OGI cameras integration, without testing possible benefits of other sensors like TDLAS, which are left for future research.

Methodology

Sensor fusion for methane detection and quantification requires integration of different sensors at various levels: hardware, electrical and software.

At the hardware level, integrating multiple sensing modalities—such as two Optical Gas Imaging (OGI) cameras onto a robotic platform presents several engineering challenges and design considerations. Each sensor type has distinct physical, operational, and environmental requirements that must be reconciled in the system architecture.

For dual OGI cameras, careful alignment and mounting are required to ensure overlapping fields of view or stereoscopic coverage, depending on the fusion strategy. Mechanical stability is critical to maintain calibration and prevent parallax errors during movement. Consideration must also be given to each camera's lens characteristics, spectral response (typically in the mid-wave infrared), and thermal management, as OGI systems can be sensitive to temperature fluctuations.

Furthermore, each sensor introduces specific power, communication, and environmental constraints. OGI cameras typically require high-bandwidth video output and low-latency synchronization. The robot's payload, power budget, and cooling capacity must accommodate all sensors without compromising mobility or endurance. The design must also account for electromagnetic compatibility (EMC), potential cross-sensor interference, and environmental sealing to protect against dust, moisture, or explosive atmospheres in field deployments. Maintaining the explosive atmospheres (ATEX) certification while integrating ATEX sensors is complex problem that requires careful considerations.

While the hardware integration of advanced methane sensing systems can be mechanically complex, the electrical integration is often comparatively straightforward—particularly when these sensors are designed as standalone units. In many non-robotic deployments, each sensor operates with its own independent power supply and internal data handling, requiring minimal coordination at the electrical level.

However, when transitioning to a robotic platform, some consolidation and coordination of electrical systems become necessary. One of the primary concerns is power distribution. The platform must either provide a centralized power supply that meets the voltage and current requirements of all sensors or accommodate each sensor's native power supply modules, ensuring isolation and stability. For example, high-power OGI cameras may require 12V or 24V DC with peak current draw during startup or image processing. The electrical design must prevent brownouts or noise coupling, especially if sensors are sensitive to voltage ripple or power transients.

Signal interfacing is another consideration, though often limited in scope. If the sensors output digital signals (e.g., Ethernet, RS-232/485, or USB), the integration primarily involves routing and managing data lines to the robot's onboard compute unit. In some cases, galvanic isolation may be needed to prevent ground loops or signal degradation. Analog output (such as from older or simplified gas sensors) may necessitate ADC channels with appropriate filtering and impedance matching, but this is less common with modern industrial-grade OGI systems.

Software integration is the critical layer in achieving effective sensor fusion for methane detection and quantification. Unlike electrical integration—which may be minimal—software integration is where data streams are ingested, synchronized, interpreted, and ultimately fused into actionable outputs. It encompasses multiple tasks including device interfacing, data acquisition, time synchronization, preprocessing, sensor fusion, and visualization or control logic.

The first challenge lies in interfacing with the sensors. OGI cameras often provide thermal video streams or frame captures over GigE Vision, USB, or proprietary SDKs. The software must be able to concurrently manage these disparate I/O channels, handle exceptions (e.g., dropped frames, communication timeouts), and buffer or align data based on timestamps. Middleware such as ROS (Robot Operating System), LabVIEW, or custom Python/C++ pipelines may be employed to manage this complexity in a modular and scalable way.

Time synchronization is another key aspect, particularly when aiming to correlate methane plume imagery from OGI. Even if sensors are not hardware-synchronized, software-level timestamping (using system clocks or synchronized NTP/PTP time) enables temporal alignment and joint interpretation. For applications requiring real-time response—such as autonomous gas source localization or leak quantification—low-latency, deterministic processing pipelines may be necessary.

For quantitative methane measurement using two OGI cameras we propose a stereoscopic quantitative optical gas imaging (sQOGI) pipeline that combines radiometric inversion with stereo depth reconstruction.

The stereoscopic quantitative-OGI (sQOGI) workflow begins with a dual-camera calibration phase: each mid-wave-IR camera is radiometrically referenced against a two-point black-body and geometrically co-calibrated with a heated-chessboard target to recover intrinsic matrices, lens distortion, and a 6-DoF stereo baseline. This step lets raw pixel DN values be linearised and converted to path-integrated column density ($\text{ppm} \cdot \text{m}$) through Beer–Lambert inversion, using the known methane absorption band and the camera's stated Noise-Equivalent-Concentration-Length (NECL) curve. (C. Ilonze et al. (2024))

Real-time acquisition uses hardware-triggered frame grabbing at 30 Hz so that both cameras share a common clock. Incoming frames are first background-subtracted with an adaptive temporal median; a light U-Net variant then refines a plume mask and suppresses thermal clutter. Only masked regions are passed to the depth engine, where a thermal-aware stereo network—adapted from ThermoStereoRT's attention-based disparity estimator (A. Hu et al. (2025))—computes dense disparity. Rectified disparities are triangulated to produce a 3-D point cloud of plume voxels; each 3-D ray also carries its measured column density.

To turn those line-integrated densities into voxel concentrations, we run an iterative algebraic reconstruction (SART) solver on a CUDA kernel. Starting from a homogeneous prior, SART repeatedly projects the trial 3-D field into both camera views and updates the voxel grid until the reprojection error falls below the image-domain noise floor. Dual-view tomography of IR plumes has shown that two properly spaced viewpoints ($10\text{--}25^\circ$) are sufficient for stable inversions in the absence of strong occlusion (R. Kang

et al. (2022)), and our solver follows the update strategy demonstrated in X. Watremez et al. (2016)'s real-time 3-camera system.

Instead of relying on an external anemometer, the pipeline derives a wind proxy directly from the two-camera video stream. Consecutive, background-subtracted frames are fed to a pyramidal Lucas–Kanade optical-flow solver that is restricted to the validated plume mask; the resulting dense pixel-velocity field is then filtered with RANSAC to keep the dominant motion component. Using the stereo-recovered depth map, each pixel displacement Δp (pixels $\cdot$ frame⁻¹) is converted to a physical speed $v = \Delta p \cdot (\text{px size at depth}) / \Delta t$, yielding a 3-D plume-advection vector for every voxel. An integration plane is positioned one plume-width downwind along this vector; voxel concentrations are multiplied by local air density and the normal component of the data-derived velocity field, and the product is surface-integrated to produce instantaneous mass-flux (g s⁻¹).

Table 1—Overview of the proposed Stereoscopic Quantitative Optical Gas Imaging algorithm for emission quantification

ALGORITHM 1: STEREOSCOPIC QUANTITATIVE OPTICAL GAS IMAGING (sQOGI)
Calibration Phase:
<i>Input:</i> Two mid-wave-IR cameras, two-point black-body, heated-chessboard target
<i>Output:</i> Intrinsic matrices, lens distortion, 6-DoF stereo baseline
Real-Time Acquisition:
<i>Input:</i> Hardware-triggered frame grabbing at 30 Hz
<i>Output:</i> 3-D point cloud of plume voxels with measured column density
Background-subtract incoming frames with an adaptive temporal median
Refine plume mask and suppress thermal clutter using a light U-Net
Pass masked regions to the depth engine
Triangulate rectified disparities to produce a 3-D point cloud of plume voxels
Algebraic Reconstruction:
<i>Input:</i> 3-D point cloud of plume voxels
<i>Output:</i> Voxel concentrations
Initialize a homogeneous prior
Run an iterative algebraic reconstruction (SART) solver
Wind Proxy Derivation:
<i>Input:</i> Two-camera video stream
<i>Output:</i> 3-D plume-advection vector for every voxel
Feed consecutive frames to an optical-flow algorithm
Calculate dominant motion component
Convert pixel displacement to physical speed
Estimate the mass-flux

Results and Observations

As a case study we used two OGI cameras from different manufacturer, in particular the Sensia Mileva and the Sierra-Olympia Ventus. In order to gather data, we successfully developed a novel driver to facilitate sensor integration and manage simultaneous data acquisition from two distinct OGI (Optical Gas Imaging) cameras. This driver was engineered to address the unique challenges posed by the cameras’ disparate data formats and communication protocols. By leveraging a multi-threaded architecture, we ensured real-time data streaming and processing, enabling concurrent operation without latency issues. The driver infrastructure is depicted in the following Figure 1

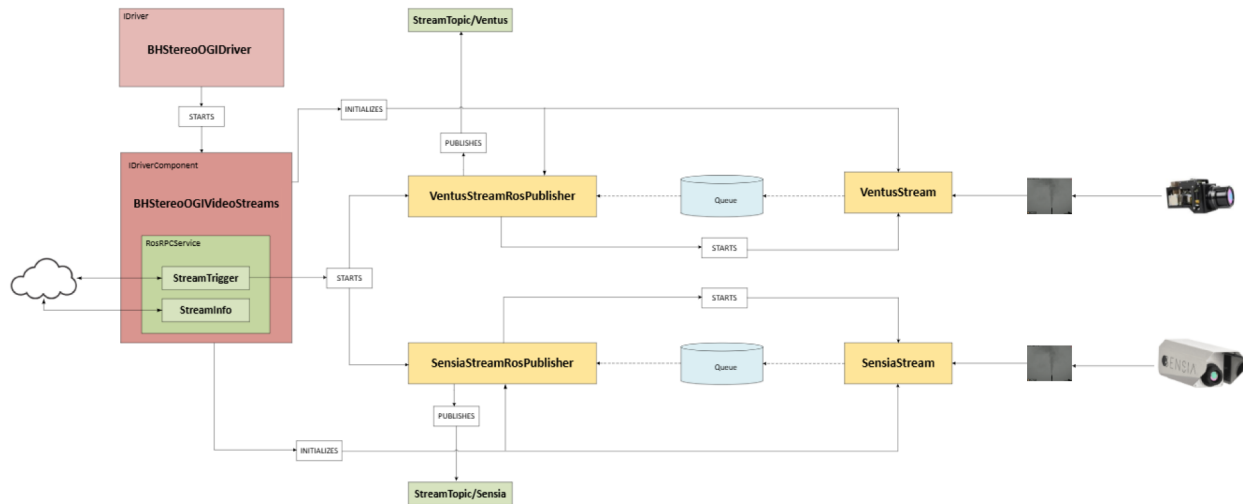


Figure 1—Infrastructure for frame acquisition from both OGI cameras and synchronization for frame superimposition, as needed from quantification algorithm

We use the Robot Operating System (ROS) infrastructure as a message broker. Each frame is paired with a timestamp and stored in a queue, then frames from different cameras are read from queue and compared to ensure that the timestamps are close enough to enable superimposition. Additionally, we implemented robust error-handling protocols to maintain data integrity and system reliability.

To integrate the Sensia Mileva with the Sierra-Olympia Ventus, a custom bracket was engineered and 3D printed (Figure 2). This bracket encases the Sensia Mileva, with a screw securing the connection through the tripod mount hole. The Sierra-Olympia Ventus is then mounted on top and fastened using the screw holes at its base. For further research, this test-ready configuration includes additional space behind the Ventus to accommodate connections for measuring lasers, digital levels, or other instruments.

A Python script has been created to simultaneously view and record videos from both cameras. This functionality is encapsulated within a class named ‘CameraStream’, which uses threading to handle the video streams. Frames are captured using the OpenCV library, timestamped, and queued for processing. They are then retrieved, written to an output file, and displayed on the screens. The output files are named based on the respective camera and the start time of the recording.

To enable both devices to stream video over Ethernet, one of them will require an Ethernet to USB adapter. Proper configuration of IP addresses for both connections is crucial to ensure the computer can access the streams.

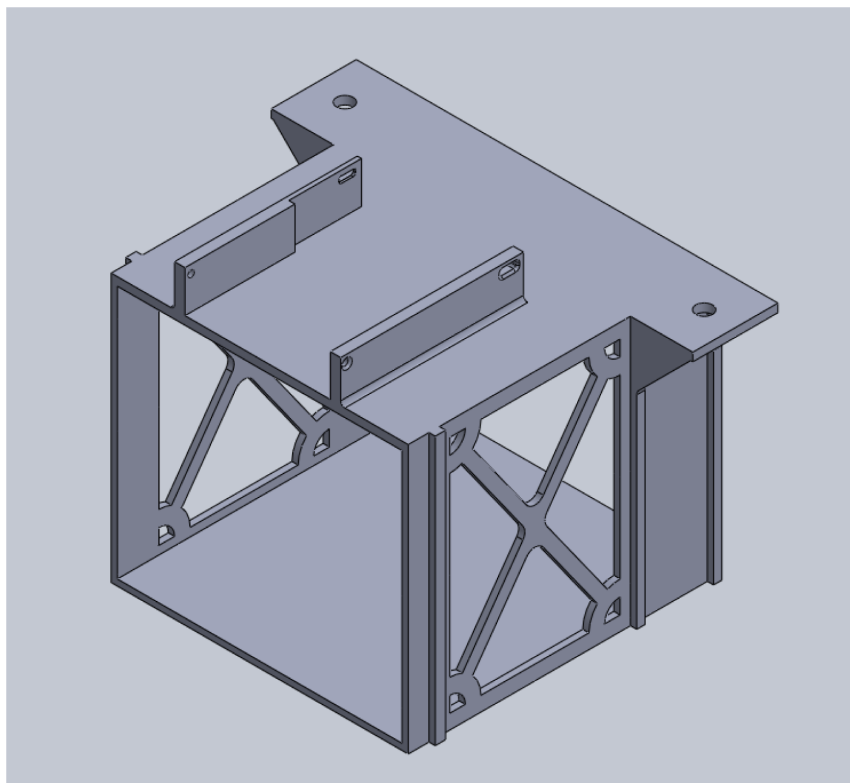


Figure 2—CAD view of a custom bracket engineered to integrate Sierra-Olympia Ventus and Sensia Mileva on a generic platform.

Finally, the hardware integration was tested on our lab for acquisition, in [Figure 3](#) we show two superimposable frames captured from the two OGI cameras.



Figure 3—Two superimposable frames captured from the two OGI cameras and synchronized

On the other hand, the algorithm for emission quantification was deployed as a Python script and is ready for testing. We leave the testing phase and result report for future research, using the already tested hardware infrastructure. In conclusion, OGI is a promising technology for emission monitoring and quantification but still lacks in precision and robustness in an industrial scenario. To tackle the problem, sensor fusion is a valid approach to increase the confidence and accuracy of the measurement, as well as its robustness.

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